Gradient boosting for classification

Log loss, regression trees as base learners

Data: twelve training runs

Summary.
1. The model keeps a *score* F for each run. A high score means "probably diverges", a low score means "probably converges".
2. A formula turns the score into a probability, and a second formula measures how wrong that probability is (the *loss*).
3. Look at how far each probability is from the truth: the *residual*.
4. Build a tiny tree (a stump) that groups runs with similar residuals, and decide, for each group, *how much to add to its scores*.
5. Add a small fraction of that to the scores and repeat. The final score is the sum of all the small additions.

Colour key: every symbol keeps its colour in every formula

y	**the truth**	1 if the run diverged, 0 if it converged
F	**the score**	any number; the model's running total
p	**the probability**	$p = \sigma(F)$: chance of divergence
r	**the residual**	$r = y - p$: what is still unexplained
γ	**the leaf value**	the number added to the score of every run in a group
ν	**the step size**	the fraction we actually add (here 0.5)
g	**the slope**	$g = p - y = -r$: slope of the log loss in F
h	**the curvature**	$h = p(1-p)$: how strongly p reacts to F

Questions a tree can ask: rate batch warm-up ; outcomes: **D** diverged, **C** converged.

1. The problem

Classification means predicting a *class*, here one of two. A neural network was trained twelve times. For each run we know the learning rate, the batch size, whether warm-up was used, and whether training **D** *diverged* or **C** *converged*. For a new run we want to predict which, or better, to say *how likely* divergence is.

run	rate	batch	warm-up	outcome	y
1	0.3	256	yes	D	1
2	0.3	32	no	D	1
3	0.1	128	yes	D	1
4	0.1	64	no	D	1
5	0.03	32	no	D	1
6	0.01	32	no	D	1
7	0.03	256	no	C	0
8	0.03	64	yes	C	0
9	0.01	128	yes	C	0
10	0.01	32	yes	C	0
11	0.001	64	no	C	0
12	0.001	256	yes	C	0

We write $y = 1$ for **D** and $y = 0$ for **C**. Six runs diverged and six converged. A tree can ask yes/no questions about the three features; there are eight possible questions: four on the rate (thresholds $0.0055, 0.02, 0.065, 0.2$), three on the batch size ($48, 96, 192$) and one on warm-up.

2. From a score to a probability

A tree adds up numbers, and a sum of numbers can be anything: -7, 0, 3.2. A probability must lie between 0 and 1. The bridge is the **sigmoid** function:

$$p = \sigma(F) = \frac{1}{1 + e^{-F}}.$$

Here F is the **score** and p is the model's probability that the run diverges. It has three properties we use constantly:

- $F = 0$ gives $p = 0.5$: "I do not know".
- a positive score gives $p > 0.5$ (more likely **D**), a negative score gives $p < 0.5$ (more likely **C**);
- the bigger $|F|$ is, the closer p is to 1 or 0: confidence, but never exactly 0 or 1.

score F	$p = \sigma(F)$
-4	0.018
-3	0.047
-2	0.119
-1	0.269
0	0.500
$+1$	0.731
$+2$	0.881
$+3$	0.953
$+4$	0.982

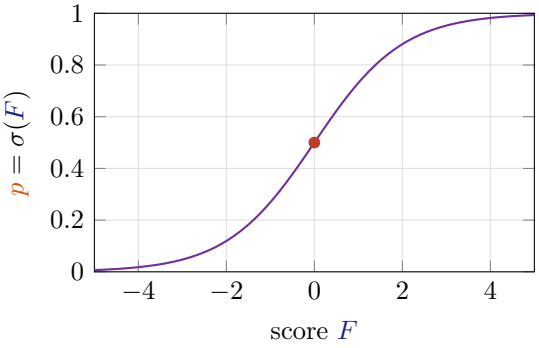


Figure 1. The sigmoid turns any score into a probability. At $F = 0$, $p = 0.5$.

Your turn 1: scores and probabilities

Compute $\sigma(1)$ and $\sigma(-1)$ using $e^1 = 2.718$. Check that they add up to 1. Then say what score would give $p = 0.9$ roughly.

Hint: $\sigma(1) = 1/(1 + e^{-1})$ *with* $e^{-1} = 0.368$. *From the table, which score gives about* 0.88 *and* 0.95*?*

Where does the model start? With no tree at all we give every run the same score $F = 0$, so $p = 0.5$ for everybody: six diverged and six converged, so "fifty-fifty" is exactly the overall rate. (If, say, 9 of 12 had diverged, the best start would be the score $\ln \frac{9}{3} = 1.10$, the *log-odds*. Here the log-odds are $\ln \frac{6}{6} = 0$.)

3. The loss: how wrong is a probability?

For one run, the **log loss** is

$$L = -\big[y \ln p + (1 - y) \ln(1 - p)\big].$$

Read it simply: if the run diverged ($y = 1$) the loss is $-\ln p$; if it converged ($y = 0$) the loss is $-\ln(1 - p)$. In both cases it is **minus the log of the probability the model gave to what actually happened**.

probability given to the truth	loss $= -\ln(\cdot)$	meaning
0.99	0.010	almost certain, right
0.90	0.105	confident, right
0.50	0.693	don't know
0.10	2.303	confident, wrong
0.01	4.605	almost certain, wrong

Being confidently wrong costs a lot (2.30 at 0.1, 4.61 at 0.01), being confidently right costs almost nothing. At the start, $p = 0.5$ for everyone, so every run has loss $-\ln 0.5 = \ln 2 = 0.693$ and the average loss is 0.693. **The aim of boosting is to bring this number down.**

Your turn 2: compute two losses

A model says $p = 0.8$ for a run. What is the loss if the run diverged ($y = 1$)? What if it converged ($y = 0$)? Use $\ln 0.8 = -0.223$ and $\ln 0.2 = -1.609$.

Hint: for $y = 1$ use $-\ln p$; for $y = 0$ use $-\ln(1 - p)$.

4. The residual: what is still unexplained

For each run, the **residual** is $r = y - p$: the truth minus the probability. A diverged run ($y = 1$) with $p = 0.5$ has $r = +0.5$: the probability should *go up*. A converged run ($y = 0$) with $p = 0.5$ has $r = -0.5$: it should *go down*. A run the model already gets right with confidence has a residual near 0.

(Why this particular quantity is the one to use is derived in Box D1: it is the negative slope of the loss with respect to the score.)

5. Round 1, step by step

1 Step 1. Compute p and r for every run. All twelve start at $F = 0$. (The column $h = p(1-p)$ is explained in Step 3.)

run	outcome	score F	p	residual r	$h = p(1-p)$	rate ≤ 0.065?
1	D	0.0	0.500	+0.500	0.25	no
2	D	0.0	0.500	+0.500	0.25	no
3	D	0.0	0.500	+0.500	0.25	no
4	D	0.0	0.500	+0.500	0.25	no
5	D	0.0	0.500	+0.500	0.25	yes
6	D	0.0	0.500	+0.500	0.25	yes
7	C	0.0	0.500	−0.500	0.25	yes
8	C	0.0	0.500	−0.500	0.25	yes
9	C	0.0	0.500	−0.500	0.25	yes
10	C	0.0	0.500	−0.500	0.25	yes
11	C	0.0	0.500	−0.500	0.25	yes
12	C	0.0	0.500	−0.500	0.25	yes

2 Step 2. Choose a question that groups the residuals. The stump should put runs with positive residuals together and runs with negative residuals together. For a yes-group L and a

no-group R:

$$\text{score} = \frac{\left(\sum_{i\in L} r_i\right)^2}{n_L} + \frac{\left(\sum_{i\in R} r_i\right)^2}{n_R}.$$

It is large when each group's residuals pull the same way. We choose the question with the largest score. (This is exactly the score of the regression document; Box D3 reviews why.) All eight questions:

question	n_L	$\sum_L r$	n_R	$\sum_R r$	score	
rate ≤ 0.065	8	-2.00	4	$+2.00$	1.500	**best**
rate ≤ 0.02	5	-1.50	7	$+1.50$	0.771	
rate ≤ 0.0055	2	-1.00	10	$+1.00$	0.600	
rate ≤ 0.2	10	-1.00	2	$+1.00$	0.600	
batch ≤ 48	4	$+1.00$	8	-1.00	0.375	
warm-up = no	6	$+1.00$	6	-1.00	0.333	
batch ≤ 192	9	$+0.50$	3	-0.50	0.111	
batch ≤ 96	7	$+0.50$	5	-0.50	0.086	

Your turn 3: score one question

Compute the score of batch ≤ 48 from the table of Step 1: which runs have batch ≤ 48 (batch 32), what do their residuals add to, and what do the other runs add to?

Hint: runs 2, 5, 6, 10 have batch 32; check your answer with the table above.

The winner is rate ≤ 0.065: its yes-group (runs 5–12) has residuals adding to -2.0 (six converged runs, two diverged), its no-group (runs 1–4, all diverged) adds to 2.0.

3 Step 3. For each group, how much should we add to the score? The stump only says *which runs belong together*. The model, however, is nothing but scores, so each group needs a number γ ("gamma") to **add to the score of every run in the group**.

Try it. Take group A (runs 5–12: two diverged, six converged) and add the same γ to all eight scores. The total loss of the eight runs:

γ added	total loss of the 8 runs	
0.0	5.545	
-0.5	4.793	
-1.0	4.506	
-1.0986	4.499	lowest
-1.5	4.611	
-2.0	5.015	
0.5	6.793	

Without any formula, we see the best γ is about -1: the loss falls while the group's score goes down (most of its runs converged) and rises again when we push too far, because the two diverged runs get hurt. For group B (runs 1–4, all diverged) the loss keeps falling as γ grows:

γ added	total loss of the 4 runs
0.0	2.773
0.5	1.896
1.0	1.253
2.0	0.508
3.0	0.194
4.0	0.073

The formula that finds this number directly.

$$\boxed{\gamma = \frac{\sum_{i\in\text{group}}(y_i - p_i)}{\sum_{i\in\text{group}} p_i(1-p_i)}}$$

- The *top*, $\sum(y-p)$, is the **total push** the group needs. Group A: two runs need $+0.5$ and six need -0.5, so -2. Group B: four runs need $+0.5$ each, so $+2$.
- The *bottom*, $\sum p(1-p)$, is the **sensitivity**: how strongly the probability reacts when the score moves. Near $p = 0.5$ one unit of score moves p by about 0.25; when p is near 0 or 1 it barely moves. Group A: $8 \times 0.25 = 2$. Group B: $4 \times 0.25 = 1$.
- Push divided by sensitivity is **the number of units of score needed to deliver the push**: group A $\gamma = \frac{-2}{2} = -1.0$, group B $\gamma = \frac{2}{1} = 2.0$.

The true lowest point for group A is $-\ln 3 = -1.099$ (table), so the formula is very close. The next part shows, step by step, where the formula comes from.

Where the formula comes from.

The question. For one group we want the number γ that makes the group's total loss smallest:

$$T(\gamma) = \sum_{i\in\text{group}} L(F_i + \gamma), \qquad L(F) = -[y \ln p + (1-y)\ln(1-p)], \quad p = \sigma(F) = \frac{1}{1+e^{-F}}.$$

Why we cannot simply solve it. In regression the total loss was exactly a parabola in γ, so its lowest point was easy to find. Here each run contributes a term like $\ln(1 + e^{-F_i - \gamma})$, and different runs have different scores F_i. Setting $T'(\gamma) = 0$ then asks us to solve an equation in which several different exponentials are added together. There is no formula for that, so we need another route.

The idea in four sentences.
1. Imagine standing on the loss curve at $\gamma = 0$, where we are now, and feeling the ground under your feet.
2. You can feel two things: how *steep* it is (the **slope**) and how quickly the steepness is *changing*, that is how much the ground curls upward like a bowl (the **curvature**).
3. With just those two numbers you can draw a parabola that hugs the curve right where you stand, and the lowest point of a parabola is easy to compute.
4. So we walk to the lowest point of that parabola and call it γ. Close to where we stand the parabola and the true curve almost coincide, so this is close to the true lowest point (Figure 2 shows how close).

Where "differentiating" enters. The slope at $\gamma = 0$ is the first derivative of the loss, and the curvature is the second derivative. The next two steps compute these two numbers; Step (iii) builds the parabola from them, and Step (iv) finds its lowest point.

Step (i): the slope. Differentiate the loss of one run with respect to its score F. Two small facts do all the work:

$$\frac{dp}{dF} = p(1-p) \qquad \text{and} \qquad \frac{dL}{dp} = -\frac{y}{p} + \frac{1-y}{1-p}.$$

The chain rule multiplies them:

$$\frac{dL}{dF} = \Big[-\frac{y}{p} + \frac{1-y}{1-p}\Big] p(1-p) = -y(1-p) + (1-y)p = p - y.$$

So the slope of the loss is $\boxed{p - y}$, which is **minus the residual**. This is why the residual $y - p$ is the right thing to fit: moving the score in the direction of the residual lowers the loss.

Step (ii): the curvature. Differentiate the slope once more, using $\frac{dp}{dF} = p(1-p)$ again:

$$\frac{d^2L}{dF^2} = \frac{d}{dF}(p - y) = \boxed{p(1-p)}.$$

It is always positive, so the loss bends upward like a bowl. It is largest (0.25) when $p = 0.5$ and nearly 0 when p is near 0 or 1.

Step (iii): the parabola. First, be clear about what is fixed and what is unknown. The scores F_i are *fixed numbers*: they are the current scores, before this round. The only unknown is γ, the amount we are about to add. For one run, write the loss as a function of γ alone, $f_i(\gamma) = L(F_i + \gamma)$. Then $\gamma = 0$ means "add nothing", so $f_i(0) = L(F_i)$ is simply *the loss the run has now.*

We expand $f_i(\gamma)$ around $\gamma = 0$, the point where we stand. The rule is $f(\gamma) \approx f(0) + f'(0)\,\gamma + \frac{1}{2} f''(0)\,\gamma^2$, and here $f_i'(0) = \frac{dL}{dF}\big|_{F_i} = p_i - y_i$ and $f_i''(0) = p_i(1-p_i)$ are the slope and curvature of Steps (i) and (ii), evaluated at the current score. This gives

$$L(F_i + \gamma) \approx L(F_i) + (p_i - y_i)\,\gamma + \tfrac{1}{2}\, p_i(1-p_i)\,\gamma^2.$$

γ itself may take any value, and the formula is a valid approximation for any γ; it is only *accurate* when γ is not far from 0, which is why the answer is close to the true bottom but not equal to it.

Add this over the runs of the group. The first terms $\sum L(F_i)$ add up to the group's current total loss, which we call $T(0)$ (it is $T(\gamma)$ with $\gamma = 0$). Writing $S = \sum(p_i - y_i)$ for the group's total slope and $H = \sum p_i(1-p_i)$ for its total curvature,

$$T(\gamma) \approx T(0) + S\,\gamma + \tfrac{1}{2}\, H\,\gamma^2.$$

Step (iv): the bottom of the parabola. Differentiate with respect to γ and set to zero:

$$S + H\gamma = 0 \quad \Longrightarrow \quad \gamma = -\frac{S}{H} = \frac{\sum(y_i - p_i)}{\sum p_i(1-p_i)}.$$

That is the formula. **Numerator** $= -S$: the total push, from the slope. **Denominator** $= H$: the sensitivity, from the curvature. A steep slope asks for a big move; a strongly curved bowl says the loss rises quickly if you overshoot, so the move must be smaller.

Check on group A (eight runs, all at $p = 0.5$, two diverged and six converged). Slope: $S = 6(0.5) + 2(-0.5) = 2$. Curvature: $H = 8(0.25) = 2$. The parabola is $T(\gamma) \approx 5.545 + 2\gamma + \gamma^2$ with its bottom at $\gamma = -S/H = -1$. The true curve is $T(\gamma) = 2\ln(1 + e^{-\gamma}) + 6\ln(1 + e^{\gamma})$, whose true bottom is -1.099. Figure 2 shows the two curves.

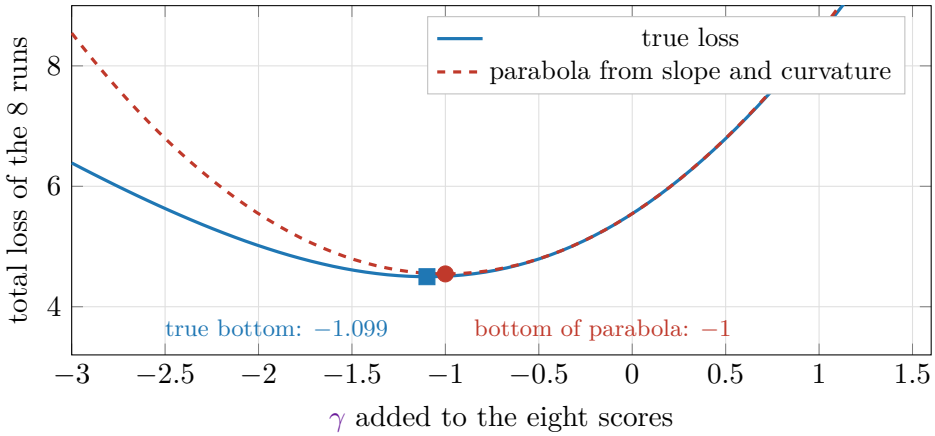


Figure 2. Group A. The parabola is built from the slope $S = 2$ and the curvature $H = 2$ at $\gamma = 0$; its bottom is almost exactly the true bottom.

Your turn 4: the leaf values

Use the formula for the two groups of the *second best* question of Step 2 (batch ≤ 48 : yes-group runs 2, 5, 6, 10; no-group the other eight runs, all at $p = 0.5$). Compute $\sum(y - p)$, $\sum p(1 - p)$ and γ for each group.

Hint: yes-group: runs 2, 5, 6 diverged, run 10 converged; every run has $p(1 - p) = 0.25$.

4 Step 4. Update the scores, with a small step. Add $\nu\gamma$ with $\nu = 0.5$: $F_i \leftarrow F_i + \nu\gamma_{\text{group}(i)}$. Why half? The sensitivity 0.25 is only true near $p = 0.5$ and falls off as p moves away, so the full step slightly overshoots; and one stump is a crude model, so we trust it only partly. After round 1:

- runs 5–12: $F = 0 + 0.5(-1.0) = -0.500$, so $p = \sigma(-0.500) = 0.378$;
- runs 1–4: $F = 0 + 0.5(2.0) = 1.000$, so $p = \sigma(1.000) = 0.731$.

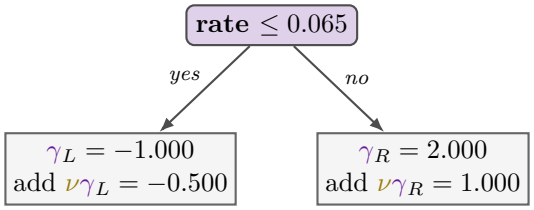


Figure 3. Tree 1: the question and, in each leaf, the number to add.

5 Step 5. Look at the mistakes again. Six of the eight runs in group A converged, and their probabilities fell from 0.5 to 0.378: better. The no-group moved from 0.5 to 0.731: better. But runs 5 and 6 diverged and sit in group A: their probability of divergence is now only 0.378, and their residual has *grown* from 0.5 to 0.622. **That is the mistake the next tree will be drawn to.** The average loss has fallen from 0.693 to 0.504.

Your turn 5: after the update

Compute the new probability p and the new residual of run 5 (diverged, in group A) and of run 7 (converged, in group A). Whose residual got worse in size, and whose better?

Hint: both have score $F = -0.5$ now, so the same p; $r = y - p$ with their own y.

6. Round 2 and round 3: the same recipe

The recipe never changes: (1) recompute $p = \sigma(F)$ and $r = y - p$ from the current scores, (2) choose the question with the largest score, (3) compute each group's γ, (4) add $\nu\gamma$. After round 1:

run	outcome	F	p	residual r	h
1	D	1.000	0.731	+0.269	0.197
2	D	1.000	0.731	+0.269	0.197
3	D	1.000	0.731	+0.269	0.197
4	D	1.000	0.731	+0.269	0.197
5	D	−0.500	0.378	+0.622	0.235
6	D	−0.500	0.378	+0.622	0.235
7	C	−0.500	0.378	−0.378	0.235
8	C	−0.500	0.378	−0.378	0.235
9	C	−0.500	0.378	−0.378	0.235
10	C	−0.500	0.378	−0.378	0.235
11	C	−0.500	0.378	−0.378	0.235
12	C	−0.500	0.378	−0.378	0.235

The residuals of runs 5 and 6 are now the largest (+0.622). Round 2 picks batch ≤ 48: yes-group (batch 32: runs 2, 5, 6, 10) has $\gamma_L = 1.260$, no-group $\gamma_R = -0.612$. Round 3 picks rate ≤ 0.065 again, with $\gamma_L = -0.629$, $\gamma_R = 1.448$: the same question can come back, because the scores have changed. The table summarises eight rounds.

round	question	γ_L	γ_R	average loss after	runs still wrong
0	(start)			0.693	1, 2, 3, 4, 5, 6
1	rate ≤ 0.065	−1.000	+2.000	0.504	5, 6
2	batch ≤ 48	+1.260	−0.612	0.438	10
3	rate ≤ 0.065	−0.629	+1.448	0.361	5, 6
4	warm-up = no	+0.800	−0.717	0.321	none
5	rate ≤ 0.065	−0.423	+1.279	0.283	none
6	batch ≤ 48	+0.949	−0.568	0.250	none
7	rate ≤ 0.02	−0.597	+0.577	0.230	none
8	batch ≤ 48	+0.603	−0.426	0.217	none

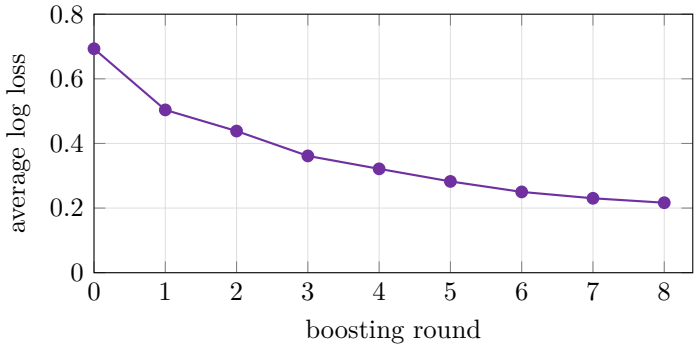


Figure 4. The average log loss, round by round. From round 4 on, no training run is on the wrong side of $p = 0.5$, yet the loss keeps falling: the probabilities keep moving towards 0 and 1.

7. The finished model, and a new run

After T rounds the **score** of a run is the sum of the numbers added by all the trees, and the probability follows from the sigmoid:

$$F(x) = \sum_{k=1}^{T} \nu\, \gamma_k(x), \qquad p(x) = \sigma\big(F(x)\big)$$

(For regression the model starts from the average; here it starts from 0, which is the log-odds of the 6/12 rate.) For a **new run B** (rate 0.2, batch 64, warm-up yes), eight rounds:

round	question	B answers	adds $\nu\gamma$	running score
1	rate ≤ 0.065	no	+1.000	+1.000
2	batch ≤ 48	no	−0.306	+0.694
3	rate ≤ 0.065	no	+0.724	+1.418
4	warm-up = no	no	−0.358	+1.059
5	rate ≤ 0.065	no	+0.639	+1.699
6	batch ≤ 48	no	−0.284	+1.415
7	rate ≤ 0.02	no	+0.289	+1.703
8	batch ≤ 48	no	−0.213	+1.490

After eight rounds $F_B = 1.490$ and $p_B = \sigma(1.490) = 0.816$: the model says B **diverges**. This is reasonable: B has a high learning rate.

8. Derive it yourself

Each formula above can be derived in a few lines. Work through the boxes in order; the full working is in Section 11.

Box D1: slope and curvature of the log loss

For one run, $L(F) = -y \ln \sigma(F) - (1-y)\ln(1-\sigma(F))$ with $\sigma(F) = \dfrac{1}{1+e^{-F}}$. (a) Show $\sigma'(F) = \sigma(1-\sigma)$. (b) Show $\dfrac{dL}{dF} = \sigma - y = p - y$. (c) Show $\dfrac{d^2L}{dF^2} = p(1-p)$. So the residual $r = y - p$ is minus the slope.

Hint: (a) differentiate $(1+e^{-F})^{-1}$. *(b) chain rule:* $\frac{dL}{dp} = -\frac{y}{p} + \frac{1-y}{1-p}$, *then multiply by* σ'. *(c) differentiate* $p - y$ *with respect to* F.

Box D2: the leaf value

Add the same γ to the score of every run in a group. Use the second-order approximation $L(F+\gamma) \approx L(F) + (p-y)\gamma + \frac{1}{2}\,p(1-p)\gamma^2$ (from D1). Sum over the runs of the group, differentiate with respect to γ, set to zero, and solve.

Hint: you will get $\sum(p_i - y_i) + \gamma \sum p_i(1-p_i) = 0$.

Box D3: the score of a question

For the first question of Section 5 (rate ≤ 0.065): $\sum_L r = -2$ with $n_L = 8$ and $\sum_R r = +2$ with $n_R = 4$. Compute the score. Then explain in a sentence why a question that mixed positive and negative residuals in each group would score close to 0.

Hint: score $= \frac{(\sum_L r)^2}{n_L} + \frac{(\sum_R r)^2}{n_R}$; *if the sums are near* 0 *the squares are near* 0.

Box D4: how good is one Newton step?

For group A (two diverged, six converged runs, all at $F = 0$) the total loss as a function of γ is $2\ln(1+e^{-\gamma}) + 6\ln(1+e^{\gamma})$. Differentiate, set to zero, and find the exact best γ. Compare with the formula's $\gamma = -1$.

Hint: derivative: $-2\,\sigma(-\gamma) + 6\,\sigma(\gamma) = 0$, *so* $\sigma(\gamma) = \frac{1}{4}$; *then* $e^{-\gamma} = 3$.

Box D5: regression is a special case

In the regression document the loss was $\frac{1}{2}(y - F)^2$. Differentiate it once and then twice with respect to F to find its slope and curvature h. The best leaf value was the average residual. Using the leaf formula of Box D2 with $p(1-p)$ replaced by the curvature, show how it contains that result.

Hint: formula: $\gamma = \frac{\sum(\text{minus slope})}{\sum(\text{curvature})}$.

9. The recipe and common confusions

Gradient boosting for classification in six lines.

0. Start with all scores $F = 0$ (the log-odds of the class rate).
1. Compute $p = \sigma(F)$ and the residual $r = y - p$ for every run.
2. Choose the question with the largest score $\frac{(\sum_L r)^2}{n_L} + \frac{(\sum_R r)^2}{n_R}$.
3. For each group set $\gamma = \sum(y - p) \,/\, \sum p(1-p)$.
4. Update $F \leftarrow F + \nu\gamma$ for every run in the group.
5. Repeat from 1. Predict $p = \sigma(F)$.

"Why not just add the average residual, like in regression?" Because probabilities saturate. A run at $p = 0.95$ needs a big change of score to move its probability a little. The denominator $\sum p(1-p)$ accounts for that: it makes the number of units of score needed larger where the probabilities are close to 0 or 1, and smaller near 0.5.

"Why a regression tree, when we are classifying?" The tree is a tool for *grouping runs* by what they still need. The residuals are numbers; the leaves hold numbers (scores to add); the class only appears at the very end, through σ.

"Why the log loss?" It punishes confident mistakes heavily, and its slope and curvature are the two simple expressions $p - y$ and $p(1-p)$.

"How does it relate to the other two?" *AdaBoost* is the same boosting idea with the exponential loss e^{-yF} and labels ± 1; its weights are each run's share of that loss. *XGBoost* is this document plus two upgrades: it chooses splits using the curvature too, and it penalises large leaf values and small gains with λ and γ.

10. Check yourself

Fill in the blanks without looking, then check against the answers below.

1. A score $F = 0$ means the probability is _____; a positive score means the run is more likely to _____.

2. The log loss of a run is minus the log of the probability the model gave to _____. At the start every run has loss _____.

3. The residual of a run is $r =$ _____. At $p = 0.5$ a diverged run has $r =$ _____ and a converged run $r =$ _____.

4. The question with the largest value of $\frac{(\sum_L r)^2}{n_L} + \frac{(\sum_R r)^2}{n_R}$ is chosen. In round 1 it is rate $\leq$ _____, with score _____.

5. The leaf value is $\gamma = \sum(y - p) \,/\, \sum$ _____. The top is the total _____ the group needs; the bottom is its _____.

6. Round 1: $\gamma_L = \frac{\quad}{\quad} =$ _____ (runs 5–12) and $\gamma_R = \frac{\quad}{\quad} =$ _____ (runs 1–4).

7. With $\nu = 0.5$ the scores become _____ (runs 5–12) and _____ (runs 1–4), so run 5 has $p =$ _____.

8. After round 1, run 5's residual has _____ (grown / shrunk) from 0.5 to _____, and the average loss has fallen from 0.693 to _____.

9. The slope of the log loss with respect to the score is _____, and its curvature is _____.

10. The exact best γ for group A is $-\ln 3 =$ _____; the formula gives _____.

11. For squared error (regression) the curvature is ____, so the leaf value becomes the _____ residual.

12. The final score of a run is the _____ of $\nu\gamma$ over all rounds, and the probability is σ of it.

Answers

1. 0.5; diverge

2. what actually happened (the truth); $\ln 2 = 0.693$

3. $y - p$; $+0.5$; -0.5

4. 0.065; 1.500

5. $p(1-p)$; push; sensitivity (how strongly p reacts to the score)

6. $-2/2 = -1$; $2/1 = 2$

7. -0.5; 1.0; 0.378

8. grown; 0.622; 0.504

9. $p - y$; $p(1-p)$

10. -1.099; -1

11. 1; average

12. sum

11. Solutions to the boxes

Your turn 1

$\sigma(1) = \dfrac{1}{1+0.368} = 0.731$ and $\sigma(-1) = \dfrac{1}{1+2.718} = 0.269$; $0.731 + 0.269 = 1$ (in general $\sigma(-F) = 1 - \sigma(F)$). A probability of 0.9 needs a score of about 2.2 (table: $F = 2$ gives 0.88, $F = 3$ gives 0.95).

Your turn 2

$y = 1$: $L = -\ln 0.8 = 0.223$ (small: it was right and fairly confident). $y = 0$: $L = -\ln(1 - 0.8) = -\ln 0.2 = 1.609$ (large: wrong and fairly confident).

Your turn 3

Runs 2, 5, 6, 10 answer yes: residuals $+0.5, +0.5, +0.5, -0.5$, sum $+1$, $n_L = 4$. The other eight runs: sum -1, $n_R = 8$. Score $= \dfrac{1}{4} + \dfrac{1}{8} = 0.375$, as in the table. Much smaller than 1.5 of the best question: its groups mix the two signs.

Your turn 4

Yes-group (runs 2, 5, 6, 10): $\sum(y - p) = 3(0.5) + (-0.5) = 1.00$; $\sum p(1-p) = 4(0.25) = 1.0$; $\gamma_L = 1.00/1.0 = 1.000$. No-group (the other eight runs, three diverged and five converged: runs 1, 3, 4 diverged): $\sum(y - p) = -1.00$; $\sum p(1-p) = 8(0.25) = 2.0$; $\gamma_R = -1.00/2.0 = -0.500$.

Your turn 5

Both runs have $F = -0.5$, so $p = \sigma(-0.5) = 0.378$. Run 5 (diverged): $r = 1 - 0.378 = 0.622$, worse than before in size ($0.5 \to 0.622$). Run 7 (converged): $r = 0 - 0.378 = -0.378$, better in size ($|r|$ fell from 0.5 to 0.378).

Box D1

(a) $\sigma = (1+e^{-F})^{-1}$, so $\sigma' = e^{-F}(1+e^{-F})^{-2} = \dfrac{1}{1+e^{-F}} \cdot \dfrac{e^{-F}}{1+e^{-F}} = \sigma(1-\sigma)$. (b) $\dfrac{dL}{dp} = -\dfrac{y}{p} + \dfrac{1-y}{1-p}$, so $\dfrac{dL}{dF} = \left[-\dfrac{y}{p} + \dfrac{1-y}{1-p}\right] p(1-p) = -y(1-p) + (1-y)p = p - y$. (c) $\dfrac{d}{dF}(p - y) = \sigma' = p(1-p)$. So the slope is $p - y = -r$: the residual is the *negative* slope, and the loss falls when the score moves in the direction of the residual.

Box D2

Summing $L(F_i + \gamma) \approx L(F_i) + (p_i - y_i)\gamma + \frac{1}{2}p_i(1-p_i)\gamma^2$ over the group gives a parabola in γ with minimum where the derivative vanishes: $\sum(p_i - y_i) + \gamma \sum p_i(1-p_i) = 0$, hence $\gamma = \dfrac{\sum(y_i - p_i)}{\sum p_i(1-p_i)}$.

Box D3

Score $= \dfrac{(-2)^2}{8} + \dfrac{2^2}{4} = 0.5 + 1 = 1.5$. If the residuals in a group have mixed signs, they cancel in the sum, so the squared sum is near 0 and the question scores low. (Derivation of the score: in the regression document, Boxes D3 and D4: it is the fall in the squared error when each leaf is set to its mean residual.)

Box D4

$\dfrac{d}{d\gamma}\left[2\ln(1+e^{-\gamma}) + 6\ln(1+e^{\gamma})\right] = -2\dfrac{e^{-\gamma}}{1+e^{-\gamma}} + 6\dfrac{e^{\gamma}}{1+e^{\gamma}} = -2\sigma(-\gamma) + 6\sigma(\gamma) = 0$. Using $\sigma(-\gamma) = 1 - \sigma(\gamma)$: $-2 + 8\sigma(\gamma) = 0$, $\sigma(\gamma) = \frac{1}{4}$, so $\dfrac{1}{1+e^{-\gamma}} = \dfrac{1}{4}$, $e^{-\gamma} = 3$, $\gamma = -\ln 3 = -1.099$. The formula's -1 is within 0.1 of it: one Newton step is close.

Box D5

Differentiating, the slope is $\dfrac{d}{dF}\frac{1}{2}(y-F)^2 = -(y-F) = -r$, and the curvature is $h = \dfrac{d}{dF}\left[-(y-F)\right] = 1$. Minus the slope is r, so $\gamma = \dfrac{\sum_{\text{leaf}} r_i}{\sum_{\text{leaf}} 1} = \dfrac{\sum r_i}{n} = \bar{r}$: the average residual. For classification the minus slope is $y - p$ and the curvature is $p(1-p)$, which gives the formula of this document.